

Evidence Detection is an NLU task which aims to detect whether a claim is entailed by some evidence. Utilising two approaches, a traditional Machine Learning method and a Deep Learning-based approach, we aim to classify previously unseen claim-evidence pairs to evaluate our implementation.

- **Dataset:** 23K labelled claim-evidence pairs; 3 columns containing claims, pieces of evidence, and labels.
- **Preprocessing:** removal of stop words and special characters, followed by tokenization and lemmatization; select sentences with the lowest Levenshtein distance [3].

Upon evaluation, it can be discerned that both models perform better than the given baseline implementations. The deep learning method's performance is attributed to the creative preprocessing and Convolutional Neural Network with Attention mechanism. The Logistic Regression model utilises TFIDF for the feature extraction and undergoes extensive hyperparameter selection to achieve the performance.

## Logistic Regression

Model:

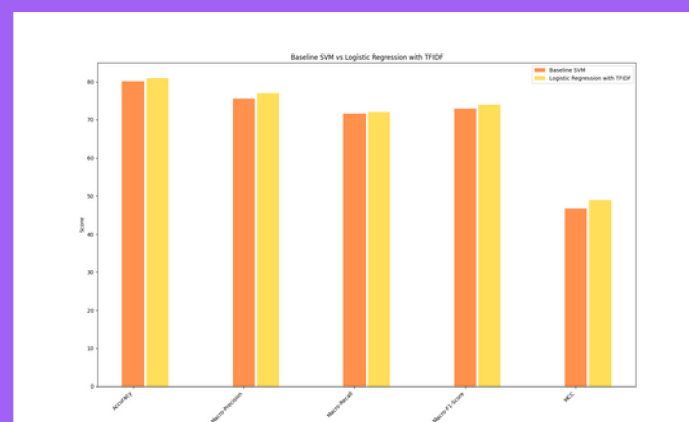
- TFIDF for feature extraction
- Robust Scaler utilized for feature normalization

Tune:

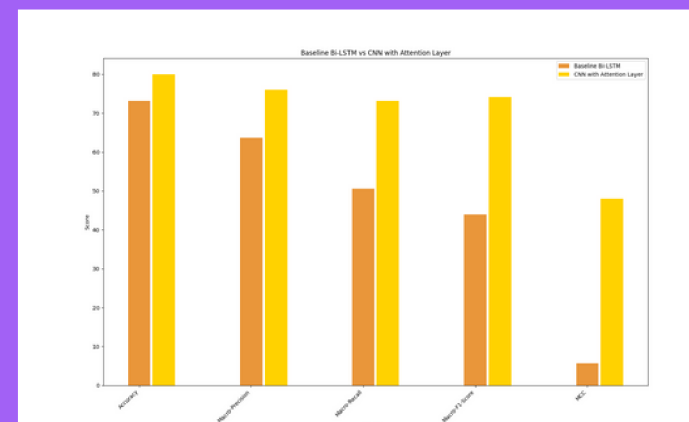
- Grid Search Cross Validation with 5 splits
- Tolerance ('tol') options - 0.0001, 0.001, 0.01
- Regularization ('C') options - 0.1, 1, 10
- Algorithm ('solver') options - 'newton-cholesky', 'lbfgs', 'liblinear', 'saga'

Optimal parameters:

- tol = 0.01
- C = 1
- solver = newton-cholesky



The proposed **Logistic Regression (LR)** performs **slightly better** in all metrics compared to the **Baseline SVM**



As evidenced above, the proposed **CNN with Attention Layer** is **significantly better** than the **Baseline Bi-LSTM**

## Results and Conclusion

| Classification Report    |          |                 |              |                |       |
|--------------------------|----------|-----------------|--------------|----------------|-------|
| Model                    | Accuracy | Macro-Precision | Macro-Recall | Macro-F1-Score | MCC   |
| Logistic Regression      | 81%      | 76%             | 72%          | 74%            | 48.6% |
| CNN with Attention Layer | 80%      | 75%             | 73%          | 74%            | 48%   |

- The LR model may be better at capturing the important features of the task based on the higher scores.
- The task may not be complex enough to warrant the use of a more complex model like CNN.
- Overall, both models achieved impressive results in determining if the evidence supports the claim.

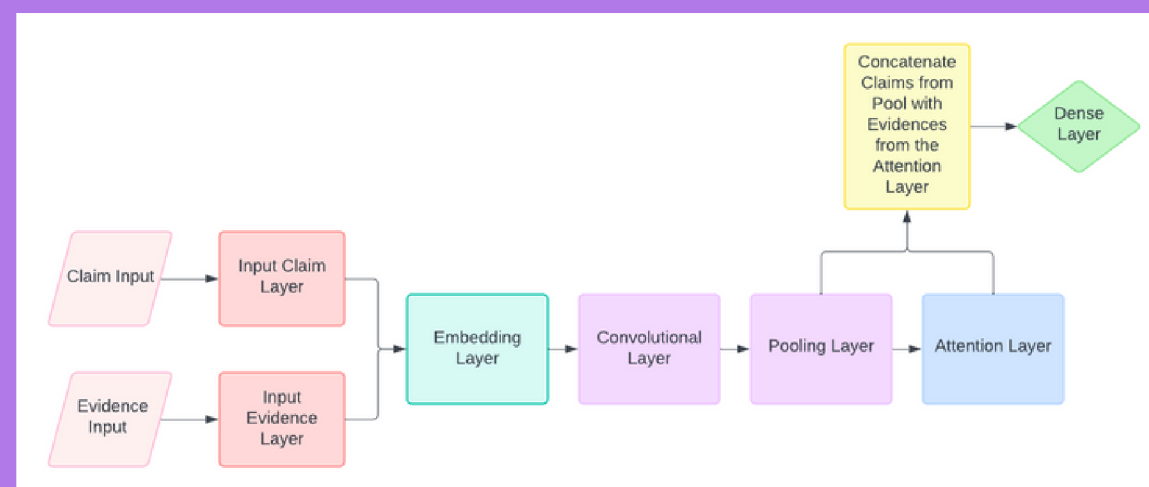
## CNN with Attention Layer

Model:

- Input Layer
- Embedding Layer
- Convolutional 1D Layer
- Global Max Pooling 1D Layer
- Attention Layer
- Concatenate Layer (Pooling claims with Attention Evidence)
- Dense Layer (Output)

Hyperparameters:

- Batch size - 64
- Validation batch size - 32



## Future Steps

- Feature Extraction:
  - word2vec or doc2vec [2]
  - Entity-Realtion-Entity [1]
- CNN-A-LSTM
- Introduce more hidden layers

## Bibliography

- [1] Jurafsky, D. and Martin, J.H. (2024). Word Senses and WordNet. [online] Available at: <https://web.stanford.edu/~jurafsky/slp3/19.pdf>.
- [2] Lee, M. (2014). Gensim: topic modelling for humans. [online] radimrehurek.com. Available at: [https://radimrehurek.com/gensim/auto\\_examples/tutorials/run\\_doc2vec\\_lee.html](https://radimrehurek.com/gensim/auto_examples/tutorials/run_doc2vec_lee.html).
- [3] Sathe, A. and Park, J. (2021). Automatic Fact-Checking with Document-level Annotations using BERT and Multiple Instance Learning. [online] ACLWeb. doi:<https://doi.org/10.18653/v1/2021.fever-1.11>.